Problem Statement

Video-Based Emotion Recognition System Using Deep Learning Technique

Human emotions play a crucial role in communication, influencing behavior, decision-making, and social interactions. The ability to automatically recognize emotions has gained significant importance in recent years, with applications in healthcare, education, customer service, virtual assistants, and human-computer interaction. While existing techniques can effectively recognize emotions from single images or simple facial expressions, accurately identifying emotions in real-world video scenarios remains a challenging task.

In video-based environments, especially those involving multiple individuals, emotions are dynamic and continuously evolve over time. Facial expressions can vary due to changes in lighting conditions, camera angles, head movements, and interactions among participants. These variations make it difficult for traditional machine learning approaches to capture both the spatial features of facial expressions and the temporal progression of emotions across video frames. Moreover, many existing systems focus on static images or short video clips, limiting their ability to understand the full context of emotional expressions in continuous video sequences.

The MELD (Multimodal EmotionLines Dataset) provides a comprehensive dataset for studying emotion recognition in conversational videos. It contains numerous video clips with annotated emotion labels, offering a realistic benchmark for research. However, developing a model that can effectively learn both facial features and temporal dependencies remains a significant challenge.

This project aims to address these challenges by developing a video-based emotion recognition system using deep learning techniques. The proposed approach utilizes a Convolutional Neural Network (CNN) to extract meaningful spatial features from individual video frames. These features are then fed into a Bidirectional Long Short-Term Memory (BiLSTM) network, which captures temporal relationships and tracks how emotions evolve over time. Finally, a classification layer predicts the corresponding emotion category, such as Happy, Sad, Angry, Fear, or Neutral.

By integrating spatial feature extraction with temporal sequence modeling, the proposed system aims to enhance the accuracy and robustness of emotion recognition. It eliminates the need for manual feature engineering and provides an end-to-end automated solution for analyzing emotions in video data. This system can contribute to the development of intelligent applications such as emotion-aware virtual assistants, adaptive learning platforms, healthcare monitoring systems, and advanced human-computer interaction technologies.

Overall, this project focuses on overcoming the limitations of existing methods by introducing a CNN–BiLSTM-based deep learning approach for effective emotion recognition in video-based environments.